Mohammed Abdul Haseeb

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Profile

Computer Science student and programming mentor with strong foundations in mathematics, programming, and problem solving. Enjoy breaking down complex concepts into simple, practical ideas and helping learners understand the connection between mathematics, logic, and programming. Have built practical projects involving simulations, algorithms, machine learning, and software systems by applying mathematical and computational concepts. Focus on hands-on learning, encouraging students to understand how and why things work rather than simply memorising formulas or syntax.

Teaching & Mentoring Experience

Peer Programming Mentor University

- Helped peers understand programming fundamentals, object-oriented programming, data structures, algorithms, and problem-solving concepts.
- Explained complex technical concepts through simple examples, visual reasoning, and step-by-step problem solving.
- Assisted peers with debugging, understanding errors, and developing stronger programming fundamentals.
- Adapted explanations to the learner's existing knowledge and pace, focusing on understanding rather than memorisation.

Teaching Areas

Programming: Python, JavaScript, C/C++, Programming Fundamentals, OOP, Problem Solving, Algorithms, Web Development, Game & Simulation Development, Introductory AI/ML

Mathematics: School Mathematics, Algebra, Geometry, Trigonometry, Coordinate Geometry, Probability & Statistics, Mathematical Logic, Discrete Mathematics

Applied Learning: Connecting mathematics and logical reasoning with programming through simulations, algorithms, games, and practical projects.

Mathematics + Programming

Enjoy connecting mathematical concepts with programming to make abstract ideas practical and easier to understand. Projects include applying vector mathematics to flocking simulations, mathematical logic and Boolean expressions to a truth-table evaluator, and mathematical foundations to machine learning and engineering systems.

Education

B.Tech in Computer Science and Engineering 2023–2027
Rajiv Gandhi University of Knowledge Technologies, Basar CGPA: 8.37

Pre-University Course 2021–2023
Rajiv Gandhi University of Knowledge Technologies CGPA: 9.56

Secondary Education (SSC) 2020–2021
Silver Wings High School, Nizamabad, Telangana CGPA: 10.0

Selected Projects

Boids Flocking Simulation Live | GitHub

- Built an interactive flocking simulation featuring predator AI and steering behaviours including alignment, cohesion, and separation.

- Applied vector mathematics, spatial reasoning, OOP, and event-driven programming to simulate the behaviour of 100+ agents.
- **Tech:** HTML5 Canvas, JavaScript, CSS.

Truth Table Printer | Logical Expression Evaluator GitHub

- Developed a Python library that generates truth tables for arbitrary logical expressions using the Shunting Yard algorithm for infix-to-postfix conversion.
- Applied Boolean logic and discrete mathematics to build a practical expression evaluator supporting NOT, AND, OR, Implication, and Biconditional.
- **Tech:** Python, Shunting Yard Algorithm, CLI.

CollabSpace | Real-Time Collaborative Whiteboard Live | Frontend | Backend

- Built a collaborative workspace supporting real-time drawing, role-based permissions, and member management.
- Designed a delta-sync protocol reducing network payload by nearly 99% while combining in-memory caching with PostgreSQL auto-save for fast loading.
- **Tech:** React, Node.js, Socket.IO, PostgreSQL, Tailwind CSS, Excalidraw.

ChatApp | Real-Time Messaging Live | Frontend | Backend

- Developed a real-time messaging platform with typing indicators, online presence, and unread message tracking.
- Implemented JWT authentication with token versioning for secure server-side logout and solved message duplication using a socket-driven synchronization model.
- **Tech:** React, Express, Socket.IO, PostgreSQL, Prisma, JWT.

DeepC | Deep Learning Library in C GitHub

- Built a neural network library from scratch in C99 featuring dense layers, activation functions, SGD, Adam, Xavier initialization, and model serialization.
- Implemented the complete training pipeline with batch gradient descent, preprocessing utilities, and multiple loss functions.
- **Tech:** C99, GCC, Make, Linear Algebra.

On-Device Audio Anomaly Detector GitHub

- Developed a convolutional autoencoder on mel-spectrograms for privacy-preserving anomaly detection.
- Quantized the model to 0.33 MB TFLite with 14.6 ms inference on CPU.
- **Tech:** Python, TensorFlow/Keras, TFLite, Librosa, scikit-learn.

Technical Experience

Edge AI Intern (Team Lead) May 2026 – Jul 2026
Socio Matics Pvt Ltd

- Led a team of four to develop **Terra-Scope**, a standalone edge-AI soil analyser providing crop and fertiliser recommendations without internet connectivity.
- Architected the software and AI stack, including Modbus communication, sensor filtering, on-device Decision Tree inference (84% accuracy, <5 ms using emlearn), keypad interface, and voice synthesis.
- Implemented STCR fertiliser equations and built a voice engine using DFPlayer with 222 mapped words for improved accessibility.

Skills

Languages: C, C++, Java, Go, Python, JavaScript
Backend: Node.js, Express, REST APIs, Socket.IO, Prisma
Databases: PostgreSQL, MySQL
Frontend: React, Tailwind CSS, HTML5, Canvas API
Machine Learning: TensorFlow/Keras, TFLite, scikit-learn, emlearn
Embedded Systems: ESP32, Modbus, UART, I2C
Tools: Git, Linux, Docker, CMake, Render, Vercel, Maven
Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks

Achievements

- Secured **First Place** in two university-level coding competitions.
- Secured **12th Place** in a multi-university coding contest organised by **9am** across three institutions including RGUKT Basar.